3.1.2 Amount of substance: Sheet 2

Empirical formula and reacting masses · 19 marks · about 25 minutes

Key information

- **n = m ÷ M_r**. Number of particles = n × L (L = 6.022×10^{23} mol^{-1}).
- Empirical formula: divide each mass (or %) by A_r, then divide by the smallest; round to whole numbers.
- Reacting masses: moles of known → use the equation ratio → moles of unknown → mass.
- % by mass of an element = (A_r × number of atoms) ÷ M_r × 100.
- Density = mass ÷ volume, so mass of a liquid (g) = volume (cm^3) × density (g cm^{-3}).

Section A: Self-quiz

01 5.0 g of an oxide of molybdenum contain 4.0 g of molybdenum. What is the empirical formula of this oxide?

[1 mark]

A MoO_2
B Mo_4O_5
C Mo_2O_3
D Mo_3O_2

02 What is the empirical formula of a hydrocarbon that contains 90% carbon by mass?

[1 mark]

A C_2H_3
B C_3H_2
C C_3H_4
D C_4H_3

03 What is the percentage by mass of sodium in sodium sulfate(VI)?

[1 mark]

A 19.3
B 22.3
C 32.4
D 36.5

04 A liquid has M_r = 169.9 and density 1.75 g cm^{-3}
Calculate the number of molecules in 5.0 cm^3 of the liquid. Give your answer in standard form.
The Avogadro constant, L = 6.022×10^{23} mol^{-1}

[2 marks]

05 Complete combustion of 0.0100 mol of an organic acid produced 0.0200 mol of carbon dioxide.
The same amount of the acid required 20 cm^3 of 1.00 mol dm^{-3} NaOH(aq) for neutralisation.
Which could be the formula of the acid?

[1 mark]

A HCOOH
B CH_3COOH
C HOOCCOOH
D $HOOCCH_2COOH$

Section B: Exam questions

06 0.500 g of a hydrocarbon is analysed. The hydrocarbon contains 0.450 g of carbon.

Calculate the empirical formula of this hydrocarbon.

[3 marks]

07 Copper can be produced from rock that contains $CuFeS_2$

07.1 Balance the equations for the two stages in this process.

…$CuFeS_2$ + …O_2 + …SiO_2 → …Cu_2S + …Cu_2O + …SO_2 + …$FeSiO_3$

…Cu_2S + …Cu_2O → …Cu + …SO_2

[2 marks]

07.2 A passenger jet contains 4050 kg of copper wiring.

A rock sample contains 1.25% $CuFeS_2$ by mass.

Calculate the mass, in tonnes, of rock needed to produce enough copper wire for a passenger jet. (1 tonne = 1000 kg)

[4 marks]

08 A student determines the percentage by mass of sodium chloride in a mixture of sodium chloride and sodium iodide.

600 mg of the mixture are dissolved in water. All of the iodide ions are converted into a precipitate of silver iodide, which is filtered off, washed and dried.

The mass of the silver iodide obtained is 315 mg

Calculate the amount, in moles, of silver iodide obtained (M_r AgI = 234.8). Use this to calculate the mass, in grams, of sodium iodide in the mixture (M_r NaI = 149.9). Then calculate the percentage by mass of sodium chloride in the mixture.

[4 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Question	Marking guidance	Comments	Mark
01	**C**	Source: 7404/1 Jun 2019 Q10	1
02	**C**	Source: 7404/2 Nov 2020 Q16	1
03	**C**	Source: 7404/1 Jun 2025 Q10	1
04	M1 amount = (5.0 × 1.75) ÷ 169.9 = 0.0515 mol M2 number of molecules = M1 × 6.022 × 10^{23} = 3.1(0) – 3.13 × 10^{22}	Source: 7404/2 Jun 2022 Q05.2 (compound named as 1-iodopropane in the original) Allow ECF from M1 to M2 M2 must be in standard form (at least 2 sf)	2
05	**C**	Source: 7405/3 Jun 2022 Q07	1
06	M1 mass H = 0.05 g M2 C = 0.45 ÷ 12(.0) = 0.0375, H = 0.05 ÷ 1(.0) = 0.05 M3 C_3H_4	Source: 7404/2 Jun 2024 Q01.2 Allow ECF from M1 to M2, and from M2 (if an attempt at moles) to M3 Alternative M1/M2: 10% H, 90% C; C = 90 ÷ 12 = 7.5, H = 10 ÷ 1 = 10	3
07.1	$4CuFeS_2 + 9½O_2 + 4SiO_2 \rightarrow Cu_2S + Cu_2O + 7SO_2 + 4FeSiO_3$ $Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2$	Source: 7404/1 Jun 2017 Q06.1 and Q06.3 Allow multiples	2
07.2	M1 Mol Cu = 4050 × 1000 ÷ 63.5 (= 63 780) M2 Mass $CuFeS_2$ = 63 780 × 183.5 (= 1.17 × 10^7 g) M3 Mass ore = 1.17 × 10^7 × 100 ÷ 1.25 M4 Mass ore = 936 tonnes (allow 936 – 937)	Alternative: M1 % Cu in $CuFeS_2$ = (63.5 ÷ 183.5) × 100 = 34.6%; M2 % Cu in rock = 34.6 × 1.25 ÷ 100 = 0.4325%; M3 mass of rock = 4050 × 100 ÷ 0.4325 = 936 416 kg; M4 936 tonnes	4
08	M1 n(AgI) = 0.315 ÷ 234.8 = 1.34 × 10^{-3} mol M2 mass of NaI = 1.34 × 10^{-3} × 149.9 = 0.201 g M3 mass of NaCl = 600 – 201 = 399 mg M4 % NaCl = 399 ÷ 600 × 100 = 66.5%	Source: 7404/1 Nov 2020 Q07.4–07.6 (stem shortened, parts combined) M4 allow 66.5 – 68.3 OR M3 % NaI = (M2 × 1000) ÷ 600 × 100, then M4 = 100 – M3	4